

Galileo and the Discovery of Continuous Change

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Outline

1. **Galileo's Rolling Ball Experiment** – Historical context and setup (20 min)
2. **Analyzing the Data** – First and second increments (25 min)
3. **From Discrete to Continuous** – The mathematical pattern (15 min)
4. **Physical Interpretation** – Position, velocity, and acceleration (15 min)

1 Galileo's Rolling Ball Experiment

1.1 Historical Context

The problem of motion: Before Galileo, scholars relied on Aristotle's ideas about motion—objects moved because of their “natural tendencies.” There was no mathematical framework for describing *how* things moved.

Galileo Galilei (1564–1642): Often called the “father of modern science,” Galileo insisted on *measurement* and *mathematics* as the keys to understanding nature.

His famous quote: “The book of nature is written in the language of mathematics.”

1.2 The Experiment

Setup: Galileo rolled a ball down an inclined plane (a ramp). He wanted to understand the relationship between *time* and *position*.

The challenge: Clocks in Galileo's time were not precise. He used a water clock—measuring time by the amount of water that flowed from a container.

The method: Galileo placed gates (bells or markers) along the ramp. He adjusted their positions until the ball triggered each gate at *equal time intervals*.

1.3 The Remarkable Discovery

Observation: To mark equal time intervals, the gates needed to be placed at positions:

$$1, \quad 4, \quad 9, \quad 16, \quad 25, \quad 36, \quad \dots$$

The pattern: These are **perfect squares**!

$$1^2, \quad 2^2, \quad 3^2, \quad 4^2, \quad 5^2, \quad 6^2, \quad \dots$$

Mathematical formulation: If $x(t)$ represents the position at time t , then:

$$x(t) = c \cdot t^2$$

where c is a constant depending on the steepness of the ramp.

Why this matters: Position follows a **quadratic** function of time. This was the first precise mathematical description of accelerated motion.

2 Analyzing Galileo's Data

2.1 Building a Data Table

Setup: Let's work with $x(t) = t^2$ (setting $c = 1$ for simplicity).

Computing positions:

Time t	Position $x(t) = t^2$
0	0
1	1
2	4
3	9
4	16
5	25
6	36

2.2 First Increments

Definition: The **first increment** is the change in position from one time to the next:

$$\Delta x = x(t+1) - x(t)$$

Computing first increments:

Time t	Position $x(t)$	First increment Δx
0	0	—
1	1	$1 - 0 = 1$
2	4	$4 - 1 = 3$
3	9	$9 - 4 = 5$
4	16	$16 - 9 = 7$
5	25	$25 - 16 = 9$
6	36	$36 - 25 = 11$

Pattern: The first increments are 1, 3, 5, 7, 9, 11, ... — the **odd numbers**!

2.3 Second Increments

Definition: The **second increment** is the change in the first increment:

$$\Delta^2 x = \Delta x(t+1) - \Delta x(t)$$

Computing second increments:

Time t	Position $x(t)$	First increment	Second increment
0	0	—	—
1	1	1	—
2	4	3	$3 - 1 = 2$
3	9	5	$5 - 3 = 2$
4	16	7	$7 - 5 = 2$
5	25	9	$9 - 7 = 2$
6	36	11	$11 - 9 = 2$

Key observation: The second increment is **constant**!

2.4 Why is the Second Increment Constant?

Algebraic verification: For $x(t) = t^2$, let's compute the increments symbolically.

First increment:

$$\begin{aligned}
 \Delta x &= x(t+1) - x(t) \\
 &= (t+1)^2 - t^2 \\
 &= t^2 + 2t + 1 - t^2 \\
 &= 2t + 1
 \end{aligned}$$

Check: At $t = 2$: $\Delta x = 2(2) + 1 = 5$. ✓

Second increment:

$$\begin{aligned}\Delta^2 x &= \Delta x(t+1) - \Delta x(t) \\ &= [2(t+1) + 1] - [2t + 1] \\ &= 2t + 2 + 1 - 2t - 1 \\ &= 2\end{aligned}$$

Result: The second increment is always 2, regardless of t .

3 From Discrete to Continuous

3.1 The Increment as a Function

Observation: The first increment $\Delta x = 2t + 1$ is itself a function of t .

Interpretation: As time progresses, the increment (change in position per unit time) increases. The ball moves faster and faster.

A more general increment: What if we don't wait a full unit of time?
Let h be the time interval:

$$\Delta x = x(t+h) - x(t)$$

3.2 The Rate of Change

Definition: The **average rate of change** over an interval of length h is:

$$\text{Average rate} = \frac{x(t+h) - x(t)}{h}$$

This measures "how much position changes per unit time."

For $x(t) = t^2$:

$$\begin{aligned}\frac{x(t+h) - x(t)}{h} &= \frac{(t+h)^2 - t^2}{h} \\ &= \frac{t^2 + 2th + h^2 - t^2}{h} \\ &= \frac{2th + h^2}{h} \\ &= 2t + h\end{aligned}$$

Observation: The average rate of change depends on both t (when we start) and h (how long we wait).

3.3 Making the Interval Smaller

What happens as h gets smaller?

Interval size h	Average rate at $t = 3$
1	$2(3) + 1 = 7$
0.5	$2(3) + 0.5 = 6.5$
0.1	$2(3) + 0.1 = 6.1$
0.01	$2(3) + 0.01 = 6.01$
0.001	$2(3) + 0.001 = 6.001$

Pattern: As $h \rightarrow 0$, the average rate approaches $2t = 2(3) = 6$.

Preview: This limiting value is called the **instantaneous rate of change** or **derivative**. We'll develop this fully in the next lecture.

4 Physical Interpretation

4.1 Position, Velocity, and Acceleration

Position $x(t)$: Where the object is at time t .

Velocity: The rate of change of position. Informally:

$$v(t) \approx \frac{\Delta x}{\Delta t} = \frac{x(t+h) - x(t)}{h}$$

For $x(t) = t^2$, we found this approaches $2t$ as $h \rightarrow 0$.

Acceleration: The rate of change of velocity. For our example:

$$a(t) = \text{constant} = 2$$

This is exactly what the constant second increment tells us!

4.2 The Chain of Relationships

Summary:

$$\text{Position: } x(t) = t^2$$

$$\text{Velocity: } v(t) = 2t \quad (\text{first increment pattern})$$

$$\text{Acceleration: } a(t) = 2 \quad (\text{second increment, constant})$$

Galileo's insight: Falling objects (and rolling balls on inclines) have **constant acceleration**. This was revolutionary—it meant motion could be predicted mathematically.

4.3 Connecting to Gravity

For free fall near Earth: The position of a dropped object is:

$$x(t) = \frac{1}{2}gt^2$$

where $g \approx 9.8 \text{ m/s}^2$ is the gravitational acceleration.

The derivatives:

$$\text{Position: } x(t) = \frac{1}{2}gt^2$$

$$\text{Velocity: } v(t) = gt$$

$$\text{Acceleration: } a(t) = g \quad (\text{constant!})$$

Example: A ball dropped from rest. After 2 seconds:

- Position: $x(2) = \frac{1}{2}(9.8)(4) = 19.6$ meters fallen
- Velocity: $v(2) = 9.8 \times 2 = 19.6 \text{ m/s}$
- Acceleration: $a = 9.8 \text{ m/s}^2$ (always)

5 Summary

5.1 Key Ideas from Today

1. **Galileo's discovery:** Position of a uniformly accelerating object follows $x(t) = ct^2$
2. **First increment:** $\Delta x = x(t+1) - x(t)$ measures change in position
3. **Second increment:** $\Delta^2 x$ measures change in the change—it's constant for quadratic functions
4. **Average rate of change:** $\frac{x(t+h) - x(t)}{h}$ gives velocity over an interval
5. **Physical interpretation:**
 - First increment $\leftrightarrow$ velocity
 - Second increment $\leftrightarrow$ acceleration
 - Constant second increment $\leftrightarrow$ constant acceleration (gravity)

5.2 Looking Ahead

Next lecture: We'll formalize the **instantaneous rate of change** by taking the limit as $h \rightarrow 0$. This leads to the **derivative**—one of the central concepts of calculus.

5.3 Practice Exercises

Exercise 1: Make a table of positions, first increments, and second increments for $x(t) = 2t^2$ at times $t = 0, 1, 2, 3, 4, 5$. What is the constant second increment?

Exercise 2: For $x(t) = t^3$, compute the first three first increments (at $t = 1, 2, 3$). Are they constant? What about the second increments? The third increments?

Exercise 3: Compute the average rate of change of $x(t) = t^2$ from $t = 2$ to $t = 2.5$. Then from $t = 2$ to $t = 2.1$. What value do these seem to approach?

Exercise 4: A ball is dropped from a tall building. Using $x(t) = \frac{1}{2}(9.8)t^2$:

- (a) How far has it fallen after 3 seconds?
- (b) What is its average velocity over the first 3 seconds?
- (c) What is its velocity at exactly $t = 3$ seconds? (Use the pattern we discovered.)

Worked Solutions

Exercise 1 (Solution): For $x(t) = 2t^2$, we compute positions, first increments, and second increments:

Time t	Position $x(t) = 2t^2$	First increment Δx	Second increment $\Delta^2 x$
0	0	—	—
1	2	$2 - 0 = 2$	—
2	8	$8 - 2 = 6$	$6 - 2 = 4$
3	18	$18 - 8 = 10$	$10 - 6 = 4$
4	32	$32 - 18 = 14$	$14 - 10 = 4$
5	50	$50 - 32 = 18$	$18 - 14 = 4$

The constant second increment is **4**. For $x(t) = ct^2$, the second increment is always $2c$; here $c = 2$, so $2c = 4$.

Exercise 2 (Solution): For $x(t) = t^3$, positions at $t = 0, 1, 2, 3, 4$ are 0, 1, 8, 27, 64.

First increments (at $t = 1, 2, 3$): $\Delta x(1) = 1 - 0 = 1$, $\Delta x(2) = 8 - 1 = 7$, $\Delta x(3) = 27 - 8 = 19$. They are **not constant**.

Second increments: $\Delta^2 x(2) = 7 - 1 = 6$, $\Delta^2 x(3) = 19 - 7 = 12$. They are **not constant**.

Third increments: We need one more first increment: $\Delta x(4) = 64 - 27 = 37$. So second increments are 1, 7, 19, 37, giving $\Delta^2 x = 6, 12, 18$. Then $\Delta^3 x(2) = 12 - 6 = 6$ and $\Delta^3 x(3) = 18 - 12 = 6$. The **third increments are constant** (both equal 6). For a cubic $x(t) = t^3$, the third increment is constant; this generalizes the quadratic pattern.

Exercise 3 (Solution): Average rate of change = $\frac{x(t_2) - x(t_1)}{t_2 - t_1}$.

From $t = 2$ to $t = 2.5$: $\frac{x(2.5) - x(2)}{0.5} = \frac{(2.5)^2 - 4}{0.5} = \frac{6.25 - 4}{0.5} = \frac{2.25}{0.5} = \mathbf{4.5}$.

From $t = 2$ to $t = 2.1$: $\frac{x(2.1) - x(2)}{0.1} = \frac{(2.1)^2 - 4}{0.1} = \frac{4.41 - 4}{0.1} = \frac{0.41}{0.1} = \mathbf{4.1}$.

As the interval shrinks, these average rates approach $4 = 2t|_{t=2}$, the instantaneous rate of change (derivative) at $t = 2$.

Exercise 4 (Solution): Using $x(t) = \frac{1}{2}(9.8)t^2 = 4.9t^2$ and $v(t) = 9.8t$ (velocity from the derivative of position).

(a) After 3 seconds: $x(3) = 4.9(3)^2 = 4.9 \times 9 = \mathbf{44.1}$ meters fallen.

(b) Average velocity over the first 3 seconds:

$$\frac{x(3) - x(0)}{3 - 0} = \frac{44.1 - 0}{3} = \mathbf{14.7} \text{ m/s.}$$

(c) Velocity at exactly $t = 3$: $v(3) = 9.8 \times 3 = \mathbf{29.4}$ m/s. (From the pattern: for $x(t) = \frac{1}{2}gt^2$, the instantaneous velocity is $v(t) = gt$.)